

KIRSCH, MI, USA

WMO: 725383

Lat: 41.817N

Lon: 85.433W

Elev: 925

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.3	5.7	-6.4	4.0	2.1	-0.4	5.6	8.0	24.5	27.3	21.3	29.6	5.9	230	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	89.7	73.4	86.3	72.0	83.6	70.8	76.6	84.7	74.9	82.4	73.3	80.4	9.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	128.7	80.8	72.5	124.5	79.6	71.6	120.8	78.5	40.7	84.4	38.9	82.5	37.5	80.2	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.1	18.8	16.8	-7.2	93.0	6.7	3.0	-12.0	95.2	-15.9	97.0	-19.6	98.7	-24.5	100.9		
			DB													
			WB	-7.7	79.5	6.4	2.0	-12.3	80.9	-16.1	82.1	-19.7	83.2	-24.3	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	24.4	27.3	37.0	49.0	60.1	68.9	71.9	70.1	64.1	52.1	40.4	29.8
	DBStd	19.01	11.24	10.65	11.27	9.38	8.68	6.61	5.48	5.20	7.47	9.01	9.19	10.46
	HDD50	3030	795	638	430	128	15	0	0	0	2	82	310	630
	HDD65	6273	1259	1055	869	485	203	36	8	15	105	410	737	1091
	CDD50	2924	1	2	27	98	327	566	679	624	424	149	23	4
	CDD65	693	0	0	1	6	50	151	221	174	78	12	0	0
	CDH74	5591	0	0	8	53	437	1244	1870	1247	642	90	0	0
	CDH80	1566	0	0	0	5	102	373	587	305	180	14	0	0
Wind	WSAvg	7.2	8.5	8.2	8.4	8.9	7.7	6.3	4.9	4.7	5.4	6.8	8.0	8.2
Precipitation	PrecAvg	38.5	2.4	2.0	2.4	3.6	4.2	4.1	4.0	4.0	3.3	3.2	2.9	2.5
	PrecMax	49.9	5.3	5.0	5.2	7.3	7.4	7.4	7.7	10.6	7.4	7.8	5.4	5.4
	PrecMin	28.4	0.9	0.6	0.7	0.4	1.4	1.1	2.2	1.4	1.0	0.7	0.8	0.7
	PrecStd	4.9	1.2	1.1	1.0	1.6	1.8	1.6	1.3	1.8	1.6	1.7	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	60.8	74.2	80.7	88.0	90.8	92.8	90.3	89.9	81.7	68.9	60.0
		MCWB	52.6	53.7	62.5	62.3	70.0	72.5	76.0	74.5	72.2	68.4	58.0	57.1
	2%	DB	50.3	53.6	65.9	73.4	82.2	88.2	89.7	86.3	84.7	75.5	63.1	54.4
		MCWB	47.7	49.6	57.6	59.3	66.0	71.6	75.1	72.8	70.3	63.8	55.6	52.0
	5%	DB	44.8	46.0	60.7	70.2	79.3	84.1	86.2	82.5	81.3	71.9	58.6	48.2
		MCWB	41.9	42.2	53.4	57.3	65.3	70.1	73.0	71.3	67.8	62.0	52.7	45.6
	10%	DB	38.8	41.1	54.5	64.6	74.8	81.4	82.4	81.1	77.4	66.4	54.5	43.3
		MCWB	36.5	37.5	48.5	54.0	63.5	68.9	71.3	70.3	65.9	59.4	49.1	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	56.0	63.7	66.0	72.8	77.1	79.8	77.7	75.4	70.8	60.7	57.3
		MCDB	55.5	58.1	72.8	76.2	81.8	86.2	87.8	86.7	84.4	79.8	65.7	59.4
	2%	WB	48.3	49.0	59.1	62.5	69.9	74.7	77.4	75.4	72.9	67.2	57.3	51.8
		MCDB	49.0	52.7	64.8	69.9	78.8	82.8	85.3	82.5	81.1	72.9	61.9	53.8
	5%	WB	41.9	42.8	54.3	59.3	67.7	72.8	75.5	73.9	70.7	63.6	53.5	45.9
		MCDB	44.3	46.4	58.7	66.9	75.7	80.8	83.2	80.3	77.5	68.4	57.5	47.6
	10%	WB	36.9	37.9	48.7	55.9	65.6	71.0	73.7	72.1	68.5	60.3	49.7	40.6
		MCDB	38.7	40.7	53.9	64.2	72.4	78.4	80.5	78.2	74.9	66.0	54.2	43.4
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.9	16.5	19.7	19.3	18.5	19.1	18.6	20.1	18.0	14.3	11.9
		MCDBR	16.4	18.9	23.0	25.1	23.1	21.5	21.4	20.5	23.5	23.2	19.4	16.8
		MCWBR	14.9	15.2	15.6	14.5	12.3	9.5	9.6	10.2	11.0	12.9	14.1	14.6
	5% WB	MCDBR	15.9	18.2	20.4	21.7	19.9	17.5	18.2	17.1	18.8	19.2	17.4	15.5
		MCWBR	15.2	15.6	15.2	14.9	12.2	9.5	9.9	9.8	10.4	13.1	14.2	14.6
Clear-Sky Solar Irradiance	taub	0.298	0.343	0.352	0.394	0.427	0.439	0.434	0.441	0.398	0.351	0.319	0.300	
	taud	2.372	2.232	2.333	2.240	2.190	2.198	2.198	2.222	2.191	2.295	2.423	2.463	2.415
	Ebn at Noon	269	269	283	278	270	266	266	266	260	264	266	260	258
	Edh at Noon	26	35	36	42	46	45	44	44	37	29	24	24	23
All-Sky Solar Radiation	RadAvg	458	725	1107	1465	1690	1909	1937	1683	1371	888	554	383	
	RadStd	43	103	107	133	163	124	96	89	118	89	80	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page